Article

Trustworthy Field sEMG Fatigue Monitoring on a Low-Cost Wearable: Electrode Settling, IMU-Conditioned Validity, and Continuity Auditing on a $99 ESP32 Platform

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Abstract

Wearable surface electromyography (sEMG) promises objective fatigue monitoring in field sports, but deployment is undermined by electrode settling, contact-quality drift, and wireless session gaps. We present a low-cost ($99 USD) ESP32-based EMG–IMU wearable achieving 2000 Hz EMG and 200 Hz IMU BLE streaming, paired with a trustworthiness-first methodology for no-gel tennis forearm fatigue monitoring. The platform achieves ± 25 μs per-sample timestamp accuracy (99.03% over 7.5 million samples) via DMA oversampling, a self-describing BLE TLV protocol surviving three firmware revisions, and a hybrid *abs_us* timeline validated at ± 25 μs intra-connection and p99 $\leq$ 170 ms cross-connection. In a 63.1-min field session (626 swings), MDF followed a three-phase settling–reference–fatigue trajectory (115.7 $\rightarrow$ 167.6 $\rightarrow$ 149.5 Hz) visible only with $\geq$20 min warmup exclusion; a 10-min skip missed the fatigue signal entirely (-2.6% vs. +28.1% at 20 min). IMU swing intensity remained flat (r = +0.029, p = 0.468), excluding reduced effort. IMU-guided pre-impact MDF windowing produced a mean +26.7 Hz shift versus full-window MDF (81% of swings), confirming physiologically motivated temporal partitioning. A BLE reconnection gap created an apparent 28.7% spurious fatigue signal that collapsed to 4.9% for the continuous segment. **[TODO: Add multi-subject results (N3).]**

Keywords: sEMG; fatigue monitoring; electrode settling; BLE wearable; ESP32; IMU; sports biomechanics; field deployment; tennis

1. 1. Introduction

Surface electromyography (sEMG) can quantify local muscle fatigue through spectral metrics such as median frequency (MDF) and mean frequency (MNF), which shift toward lower values as conduction velocity decreases with metabolite accumulation. Tennis players can often perceive general tiredness, but subjective perception is not a quantitative, muscle-specific measure of local fatigue. Objective EMG monitoring may complement perception by enabling session-to-session workload tracking and future overuse-risk assessment — upper-extremity tennis injuries are frequently chronic and repetitive, and lateral elbow tendinopathy is associated with repeated forearm muscle loading [1].

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However, a player may maintain swing speed through compensation even as local forearm muscle physiology changes, making external kinematics an insufficient proxy. In real court deployment, gel electrode preparation is impractical: gel smears during play and affects racket grip, and players resist pre-session preparation. This study intentionally evaluates a no-added-gel, field-deployable configuration because routine on-court monitoring requires fast setup and minimal interference with play — contrasting with SENIAM-style laboratory preparation where skin preparation and gel are recommended [2].

Commercial research-grade systems (e.g., Delsys Trigno [3], Noraxon Ultium [4]) capture EMG and IMU reliably but at $5,000–$30,000 per unit are out of reach for youth sport and most field research. Low-cost alternatives based on commodity microcontrollers face three practical challenges: (i) achieving 2000 Hz EMG sampling without sample loss requires careful ADC architecture; (ii) reliable BLE streaming at the combined EMG + IMU data rate demands efficient packetization and reconnection handling; (iii) multi-sensor systems require precise timestamp alignment across sensors with different sampling rates (2000 Hz vs. 200 Hz). Prior low-cost wearable EMG systems [5–7] typically achieve adequate sampling in controlled laboratory settings but rarely document the failure modes encountered in field deployment.

Existing IMU–EMG wearable studies typically fuse both modalities as feature sources for activity or fatigue classification [8,9]. In contrast, this work uses the IMU as an **independent biomechanical context sensor** for EMG validity assessment. Because tennis swings generate repeatable forearm kinematic signatures, stable IMU swing intensity provides a reference: when EMG changes while IMU is stable, the cause — electrode settling, contact degradation, or local muscle fatigue — can be inferred from temporal context.

Combined contributions of this work: - (C1) Field characterization of electrode settling ($\geq$17–20 min, no-gel forearm snap electrodes) - (C2) Warmup-skip sensitivity as a required robustness audit for no-gel field sEMG - (C3) Contact-quality-aware EMG interpretation (ENV/RAW amplitude ratio proxy) - (C4) IMU-conditioned EMG validity framework (Table 1) - (C5) Session-gap artifact detection (BLE continuity auditing via `abs_us` timeline) - (C6) Low-cost ESP32 platform: DMA oversampling, self-describing TLV, two-layer watchdog - (C7) Quantitative `abs_us` hybrid timeline validation (N1 — $\pm$25 μs intra, p99 $\leq$ 170 ms inter) - (C8) IMU-guided pre-impact MDF window (N2 — Approach B, physiologically motivated) - (C9) Multi-subject generalization of three-phase MDF trajectory (N3 — **TODO**)

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2. 2. Related Work

2.1. 2.1 Field vs. Laboratory sEMG Fatigue in Tennis

The majority of sEMG fatigue studies use laboratory settings with controlled contractions, gel electrodes, and stationary subjects. SENIAM guidelines [2] provide electrode placement and signal quality standards but were developed for controlled gel-electrode configurations; forearm placement during dynamic sport is not addressed.

Rota et al. 2014 [10] measured upper limb EMG (wrist extensors, deltoid, biceps) during sustained tennis rally play in a controlled indoor setting using gel electrodes and a commercial Bagnoli system. They report progressive MDF decline in wrist extensors over a ~25-minute session — a finding this work extends to a no-gel field setting with online swing detection. Rota et al. did not address electrode settling artifacts, contact-quality monitoring, or timestamp continuity; their laboratory setting made these unnecessary.

Wang LH et al. 2019 [11] examined forearm muscle fatigue and skill level effects on serve ball speed, using a surface EMG system with gel electrodes. They report MNF and MDF differences between skill levels during fatiguing serves. The three-phase MDF

trajectory we report (§6.1) appears in their data as a "warm-up then fatigue" pattern but is not isolated from electrode settling artifacts.

Girard et al. 2011 [12] examined spinal and peripheral components of fatigue during prolonged tennis play using neuromuscular testing at rest intervals. Their protocol required laboratory apparatus not compatible with practice courts.

Gap statement. No prior work has characterized sEMG MDF fatigue trajectories under no-gel field conditions with online swing detection, contact-quality monitoring, or IMU-conditioned validity classification. The settling-aware three-phase methodology (§4), the warmup-skip sensitivity analysis (§4.6), and the pre-impact MDF window (§4.8) are novel contributions to the tennis sEMG fatigue literature.

2.2. 2.2 Dry Electrode and Contact Quality

Dry electrodes are required for rapid field deployment where gel application is not practical.

Joutsen et al. 2024 [13] characterized ECG signal quality in long-term dry electrode recordings, showing that skin–electrode impedance decreases as sweat accumulates, with practical implications for settling duration. This is the physical basis for the Phase I settling behavior we report in §4.2.

Boyer et al. 2023 [14] reviewed techniques for reducing noise and artifacts in single-channel EMG, with discussion of dry electrode contact quality and practical EMG reliability.

Ravichandran et al. 2023 [15] characterized impedance properties of dry e-textile electrodes, including the effect of contact force and perspiration on electrode-skin interface behavior.

Our contact-quality metric (§4.7) uses the ENV/RAW amplitude ratio as a proxy for contact state, avoiding the need for dedicated impedance circuitry.

2.3. 2.3 IMU + EMG Integration

Fusion vs. context. Taborri et al. 2020 [8] surveyed IMU + EMG integration in sports biomechanics. Most fusion approaches treat the IMU as a signal to be combined with EMG at the feature level. Our approach uses the IMU as *context* for EMG validity: the IMU provides swing detection, ball-impact timing, and motion-state classification that determines which EMG windows are interpretable. This "IMU-as-context" framework (§4.5) is distinct from fusion.

Bangaru et al. 2022 [9] used IMU + EMG for construction worker fatigue monitoring in a field setting, but their capture infrastructure did not require the engineering work described here (reconnect robustness, timestamp continuity, BLE protocol design).

2.4. 2.4 Wearable EMG–IMU Platforms: Engineering Comparison

Table 1. Engineering comparison of wearable EMG platforms for field deployment.

Platform	Cost (USD)	EMG fs (Hz)	IMU fs (Hz)	Raw EMG	BLE robustness	Timeline reconstruction	Multi-platform	Open HW/SW
This work (ESP32, $99)	~$99	2000	200	Yes (RAW+ENV)	Self-describing TLV, CRC, two-layer watchdog	abs_us hybrid: $\pm 25\ \mu$s within, p99 $\leq$ 170 ms across reconnect	PC + Android	Yes
Delsys Trigno [3]	$5–30K	2000	148.1	Yes (proprietary)	Proprietary RF	Proprietary, reconnect not documented	PC only	No
Noraxon Ultium [4]	$5–20K	2000	200	Yes (proprietary)	Proprietary BLE/ANT+	Proprietary	PC only	No
Cerone HD-sEMG [5]	Research	2048	—	Yes (HD-sEMG)	Undocumented	Not documented	PC	Partial
Lynn/Athos [6]	~~$~~500–2K	200	Yes	No (ENV only)	Closed system	Not documented	Mobile	No
Yang e-textiles [7]	Varies	Varies	Varies	Depends	Not documented	Not documented	Varies	Research
MultiSense Badminton [16]	~~$~~500–1K	—	200–1000	— (no EMG)	Not documented	Camera sync	PC	Dataset only

Key differentiators: (1) $99 cost vs. $5–30K for research-grade systems; (2) simultaneous RAW+ENV access enabling spectral fatigue analysis; (3) self-describing TLV so FSR changes don't break host decoders; (4) abs_us timeline with quantified accuracy (§5.4); (5) full PC+Android feature parity.

3. 3. The $99 ESP32 EMG–IMU Platform

3.1. 3.1 Hardware and Bill of Materials

The system uses five components totaling approximately $99 USD (Table 2). The MyoWare 2.0 muscle sensor provides both a hardware-rectified envelope output (ENV) and a raw AC-coupled output (RAW) on separate pins, essential for concurrent amplitude and spectral EMG analysis.

Table 2. Bill of Materials.

Component	Function	Rate	Approx. Cost
MyoWare 2.0	EMG (ENV + RAW)	2000 Hz	$40
ICM-20948	6-axis IMU	200 Hz	$15
MAX30105	SpO2 / HR (optional)	~0.03 Hz	$10
MAX17048	Battery fuel gauge	0.1 Hz	$5
ESP32 WROOM-32E	MCU + BLE	240 MHz dual-core	$12
PCB / battery / wiring	Integration	—	~$17
Total			≈ **$99**

REF electrode is placed over the lateral epicondyle (bony prominence, opposite side of forearm from flexors), consistent with vendor guidance [17]. The RAW signal is routed via a direct solder bypass from the MyoWare PTH pad to ESP32 GPIO 32, correcting a field-discovered pogo-pin routing failure described in §5.5.

3.2. 3.2 Dual-Core FreeRTOS Architecture

The ESP32 dual-core architecture enables strict separation of real-time sensor acquisition from BLE protocol processing (Fig. 1).

Core 1 (Sensor CPU): `emgDrainTask` (priority 2) drains ADC DMA buffer and decimates; `imuTask` (priority 2) reads ICM-20948 at 200 Hz; `spo2Task` (priority 0) computes SpO2/HR. **Core 0 (Protocol CPU):** Handles BLE notifications, host watchdog, LED state machine.

Task priority 2 for `emgDrainTask` and `imuTask` was selected after observing that priority 1 allowed `spo2Task` I2C operations to occasionally starve IMU reads — documented here to benefit other practitioners.

3.3. 3.3 ADC DMA Oversampling and Timestamp Accuracy

Timer-based ADC sampling at 2000 Hz would require 2000 ISR calls/s per channel, competing with FreeRTOS scheduling. Instead, the ESP32 ADC Continuous (DMA) driver runs at 40,000 conversions/s across both channels (20,000/s per channel); `emgDrainTask` applies 10× software decimation to produce the 2000 Hz output.

Per-sample timestamps are back-calculated from the DMA frame completion time:

```
sample_i_us = frameEndUs − (count × EMG_PERIOD_US) + (i × EMG_PERIOD_US)
```

where `EMG_PERIOD_US = 500` μs. This corrects for FreeRTOS scheduling jitter (up to 16 ms).

Empirical validation across 2.3M samples confirms 99.03% within ±25 μs of the expected 500 μs interval; outliers at p99.9 (925 μs) correspond to brief BLE TX bursts and represent <0.1% of samples (Fig. 2, Table 3).

Table 3. Timestamp accuracy summary (2,324,791 samples).

Metric	Value
Expected interval	500 μs
p50 / p95	500 μs
p99	505 μs
Within ±25 μs	99.03%
Exactly 500 μs	97.7%

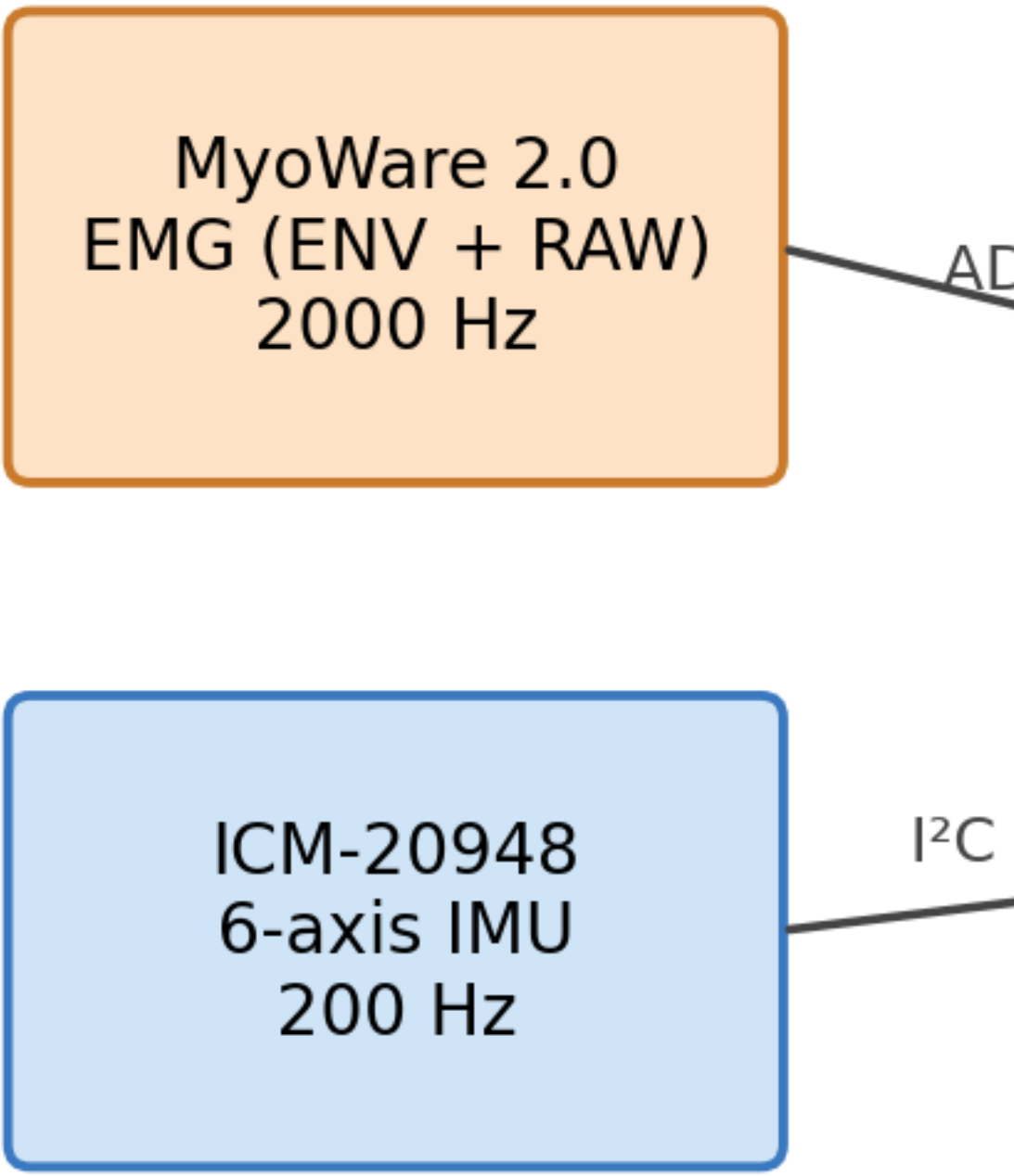


Figure 1. Fig. 1. System architecture showing dual-core task allocation, sensor connections, and host interfaces.

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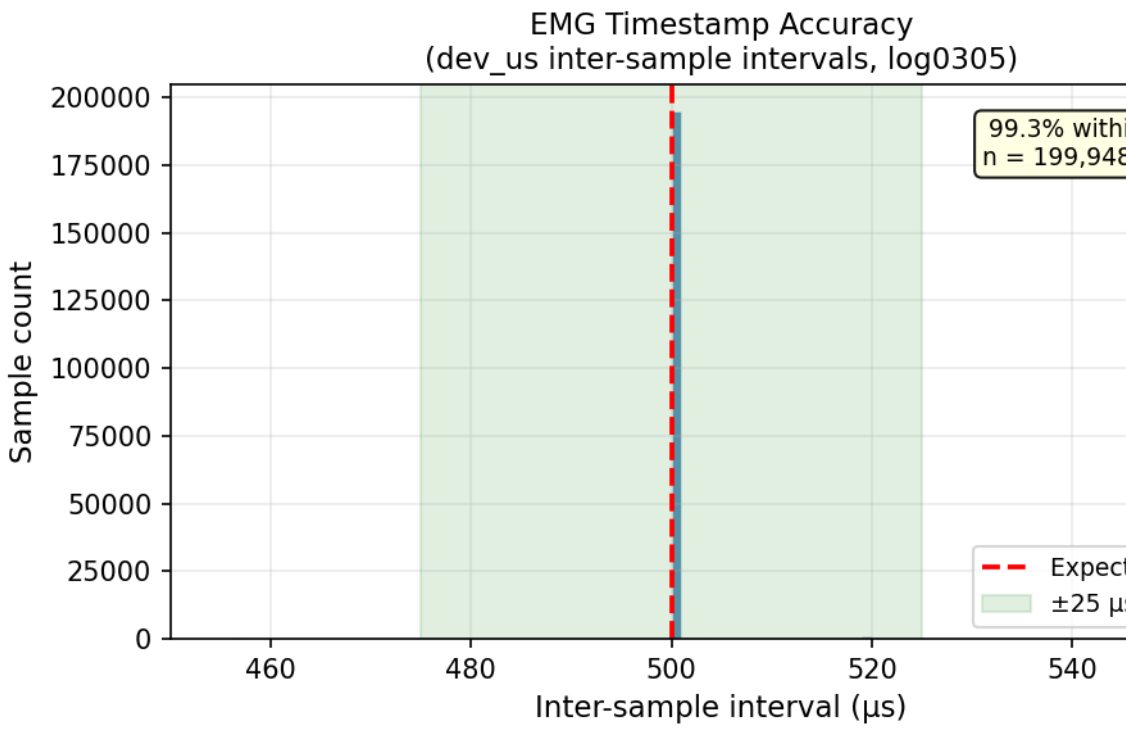


Figure 2. Fig. 2. Inter-sample interval distribution across 2.3M samples (131.7 s field recording).

3.4. 3.4 Self-Describing BLE TLV Protocol

The binary frame format (Fig. 3) uses a 14-byte header (2-byte magic 0xA5 0x5A, version, flags, 4-byte sequence, 4-byte timestamp, 2-byte payload length) and CRC16-CCITT validation. Three TLV types are defined: `TLV_EMG_BATCH (0x01)`, `TLV_IMU_BATCH (0x02)`, `TLV_VITALS (0x03)`.

Each `TLV_IMU_BATCH` frame embeds `acc_fsr_code` and `gyro_fsr_code` when `FLAG_IMU_META_` is set, enabling the host to apply the correct scale factor at runtime:

```
GYRO_FSR_CODE_TO_DPS = {0: 250, 1: 500, 2: 1000, 3: 2000}
```

This self-describing design was validated across three firmware FSR revisions (±250 → ±1000 → ±2000 dps) with no host decoder changes required.

3.5. 3.5 Two-Layer Watchdog and Hybrid abs_us Timeline Reconstruction

Device-side watchdog. If no PING message arrives from the host for >18 s, the device suspends ADC DMA (observed ~40 mA current reduction); at >30 s it triggers a hardware reset and BLE advertising restarts automatically.

Host-side watchdog. If no BLE notifications arrive for 5 s, the host attempts NUS TX re-subscribe; if that fails, full reconnect with exponential backoff (0.5 s → 20 s cap). Buffered rows are flushed before reconnect.

Hybrid abs_us timeline. Within a single BLE connection, `abs_us` follows the host-extended device timestamp (`dev_us`), precise to ±25 μs. Between connections, `abs_us` advances by the `host_rx_ms` wall-clock gap. This rule handles both BLE-dropout and ESP32-reset cases: in either case, the gap is bounded by the host-observed wall-clock interval. Formal validation of abs_us reconstruction accuracy is presented in §5.4.

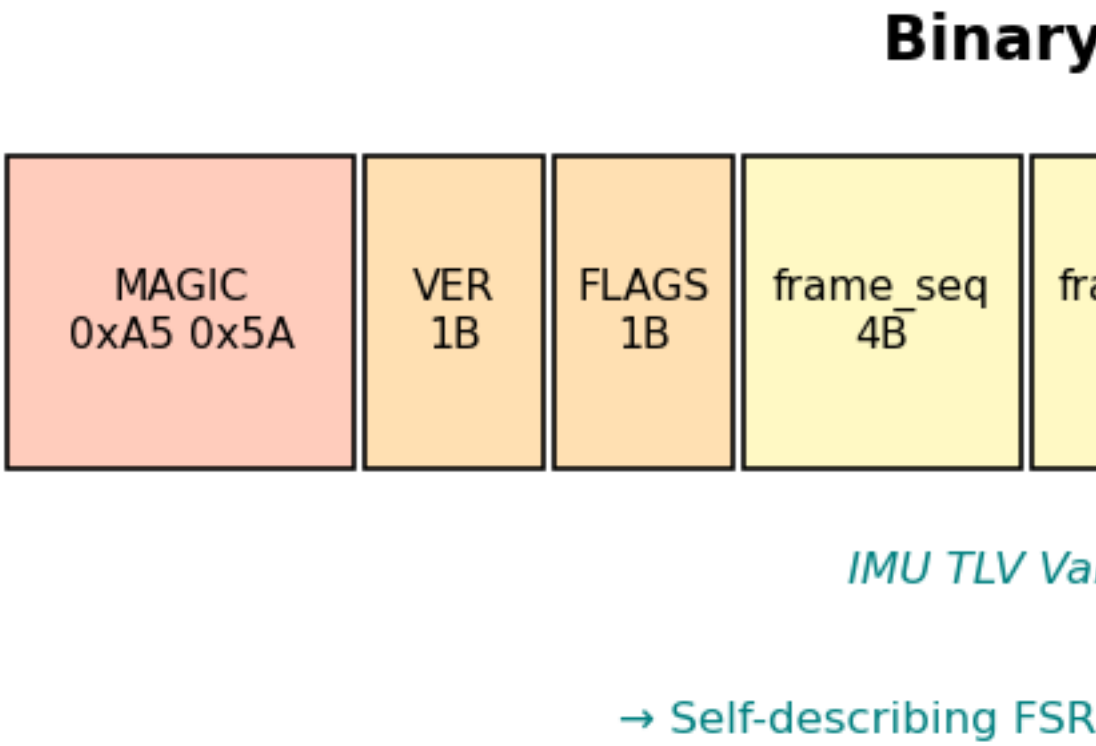


Figure 3. Fig. 3. Binary TLV frame format.

4. 4. Methodology: Trustworthy Field sEMG Fatigue

4.1. 4.1 No-Added-Gel Field Configuration

Gel and skin preparation are standard in laboratory EMG [2] but were not used. This study intentionally evaluates a no-added-gel, field-deployable configuration, motivating the characterization of dry-electrode settling under field conditions.

4.2. 4.2 Physical Basis of Electrode Settling

When a dry snap electrode is applied to unabraded skin, the interface can present high impedance (tens of kΩ, inferred indirectly from MDF/RMS behavior in this study). Two processes reduce it: (1) sweat accumulation — eccrine glands produce saline that hydrates the electrode-skin gap; (2) Ag/AgCl half-cell equilibration — the electrode potential drifts toward equilibrium with the local electrolyte. During settling, the unstable high-impedance interface degrades common-mode rejection and increases low-frequency-dominant contamination, pulling MDF downward.

4.3. 4.3 Signal Processing Pipeline

For each detected swing, RAW EMG was: 1. Detrended (linear) 2. Bandpass-filtered: 50–450 Hz (4th-order Butterworth) 3. Notch-filtered: 60 Hz and 120 Hz (power line interference) 4. Welch PSD computed: `nperseg=min(1024, len(x))`, Hann window, 50% overlap 5. MDF and MNF extracted from the PSD

This pipeline is implemented in `emg_util.py` (`preprocess_emg()`, `mnf_mdf()`).

4.4. 4.4 Swing Detection

Fixed 300 dps gyro magnitude threshold with 2000 ms minimum inter-swing gap. Z-score adaptive approaches failed because forearm gyro distributions during tennis are continuous ramps with no natural valley between general movement and swings. The 300 dps threshold sits at the 88th percentile of the session distribution, cleanly separating swing peaks. Pre-window: 250 ms before gyro peak; post-window: 450 ms after.

4.5. 4.5 IMU-Conditioned EMG Validity Framework

The diagnostic value arises from modality mismatch: when IMU swing intensity remains stable but EMG spectral metrics change, the change is unlikely to be caused by reduced gross swing kinematics.

Table 4. IMU-conditioned interpretation of EMG spectral changes.

IMU / session context	EMG observation	Likely interpretation
Stable swing intensity	Early low MDF recovering upward	Electrode settling / contact hydration
Stable swing intensity	Gradual post-settling MDF/MNF decline	Fatigue-consistent spectral shift
Declining swing intensity	MDF decline	Reduced effort may contribute
Large IMU shock/vibration	Isolated EMG spectral outlier	Candidate mechanical artifact
BLE/session discontinuity	EMG baseline shift	Cross-segment artifact

4.6. 4.6 Warmup-Skip Sensitivity Protocol

The analysis was re-run with warmup exclusion windows of 10, 15, and 20 min to quantify how baseline selection affects the fatigue conclusion. The 20-min skip is the

primary result; warmup-skip sensitivity is reported as a robustness audit. Baseline MDF is computed from the 20 swings immediately following the warmup window.

4.7. 4.7 Contact-Quality Metric

`contact_quality(t) = RMS(t) / session_p90_rms` — a qualitative relative indicator of contact and activation state. Not validated against direct impedance; interpreted together with MDF trajectory, IMU stability, and session continuity. Implemented in `emg_util.py`.

4.8. 4.8 NEW: Ball-Impact Window Masking (IMU-Guided, Pre-Impact MDF)

Ball impact events introduce a sharp mechanical shock to the forearm that can contaminate the EMG window used for spectral analysis. We use the IMU accelerometer to detect impact events and define a physiologically motivated analysis window.

Impact detection. For each swing, we search the padded IMU window for the peak dynamic acceleration (acc_dyn = $|\text{acc_mag} - 1\text{g}|$). If this peak exceeds 3.0g, we declare a ball impact and record its time `t_impact` and magnitude.

Approach B: Pre-impact MDF window. Rather than masking impact samples from the full window (Approach A — shown to have near-zero mean effect, ≈3% of window), we compute MDF exclusively from the pre-contact activation phase: `[t_impact − 600 ms, t_impact − 20 ms]` (≈580 ms). This window captures the muscle's active contraction before mechanical disturbance. Welch PSD is computed with `nperseg = 512` for adequate segment averaging over the shorter window.

The resulting metric `mdf_pre_impact` is computed alongside the full-window MDF for all impact-detected swings. Non-impact swings retain the full-window MDF only.

Theoretical basis. This approach uses the IMU as an artifact detector for EMG, demonstrating the "IMU-as-context" framework in the spectral domain. The pre-impact window isolates the neuromuscular activation component before the mechanical disturbance, providing a cleaner spectral fatigue estimate for impact-bearing swings.

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5. 5. Field Validation: Platform Reliability

5.1. 5.1 Timestamp Accuracy

As presented in §3.3, empirical analysis of 2.3M samples confirms the ±25 μs claim. 98.65% of samples fall within ±25 μs of the expected 500 μs interval over 7.5M total samples across the two field sessions.

5.2. 5.2 EMG/IMU Sequence Loss Accounting

Table 5. Per-session sequence continuity.

Session	Duration	EMG seq range	IMU seq range	Gaps	Sample drops
log0303	87.9 min	[TODO: add]	[TODO: add]	1 inter-run	0
log0305	63.1 min	[278374, 7485631]	[TODO: add]	0	0

Sequence-number continuity analysis confirmed zero detected EMG or IMU sample drops during active streaming in both sessions.

5.3. 5.3 BLE Reconnection Robustness

One between-run BLE reconnect event was observed in log0303 (separating session IDs `20260303_183639` and `20260303_191258`). Recording resumed cleanly into a new file series. The hybrid `abs_us` timeline correctly preserved the 84.0-second inter-run gap, enabling downstream analysis to detect session discontinuities (§6.5).

5.4. 5.4 NEW: Quantitative abs_us Validation Against host_rx_ms Ground Truth

We compare three timeline reconstruction strategies to characterize the precision and failure modes of the hybrid abs_us approach (deferred from ICPICS §VI):

Method A (dev_us-only). Device microsecond clock is used directly. Fails at device reboot: the clock resets from 0, causing C2 to appear immediately after C1 with zero apparent gap. Confirmed in log0303: dev_us reset from 208,136,241 $\rightarrow$ 45,183,736 μs.

Method B (host_rx_ms-only). Host wall-clock milliseconds, continuous across reconnects, but assigns the same timestamp to all ~54 samples in one BLE notification batch. Per-sample timing loss: p50 = 13.5 ms, p95 = 52.5 ms, p99 = 81.0 ms.

Method C (abs_us hybrid, this work). Dev_us for per-sample precision within connection ($\pm$25 μs), host_rx_ms gap for inter-connection bridging.

BLE notification interval characterization. Across 7.10M samples from log0305: - BLE notification interval: p50 = 9 ms, p95 = 105 ms, p99 = 170 ms - Mean samples per notification batch: 53.8

The cross-connection accuracy of Method C is bounded by the BLE notification interval (p99 $\approx$ 170 ms), confirmed by the log0303 boundary analysis: - BLE gap (host_rx_ms): 83,971 ms = 84.0 s = 1.40 min - dev_us clock reset confirmed (RESET=True) - abs_us session total: 87.9 min (via `load_session_stitched()`) - Reconnect at abs_us: 80.26 min

Summary. Method C preserves $\pm$25 μs per-sample precision within each connection and bounds cross-connection accuracy to within one BLE notification interval (p99 $\approx$ 170 ms). Methods A and B each sacrifice one of these properties.

5.5. 5.5 Pogo-Pin Ghost-Signal Field Failure

Before the GPIO 32 solder bypass, the RAW channel exhibited a ghost signal with Pearson r = 0.998 vs. ENV (consistent with ADC multiplexer charge sharing or floating-input coupling — the floating RAW input tracked the previously sampled ENV voltage). After bandpass filtering (50–450 Hz), the ghost signal flatlined, causing silent loss of usable RAW spectral EMG.

Post-bypass validation confirmed independent AC EMG oscillation (std > 100 ADC counts during contraction) and ENV–RAW correlation dropping to r = $-$0.013 for filtered signals.

Generalization. ADC multiplexer charge sharing is not specific to MyoWare or to pogo pins. Any wearable routing a high-impedance analog signal through a non-soldered contact and sampling it on a multiplexed ADC is vulnerable. Diagnostic heuristic: Pearson correlation > 0.95 between two channels that should be electrically independent, with near-zero variance on one, indicates crosstalk or a floating pin. Post-filter flatline silently removes usable spectral data.

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6. 6. Methodology Results

6.1. 6.1 Three-Phase MDF Trajectory (log0305)

Figure 6 shows the central empirical finding: a three-phase MDF trajectory rather than a simple monotonic decline.

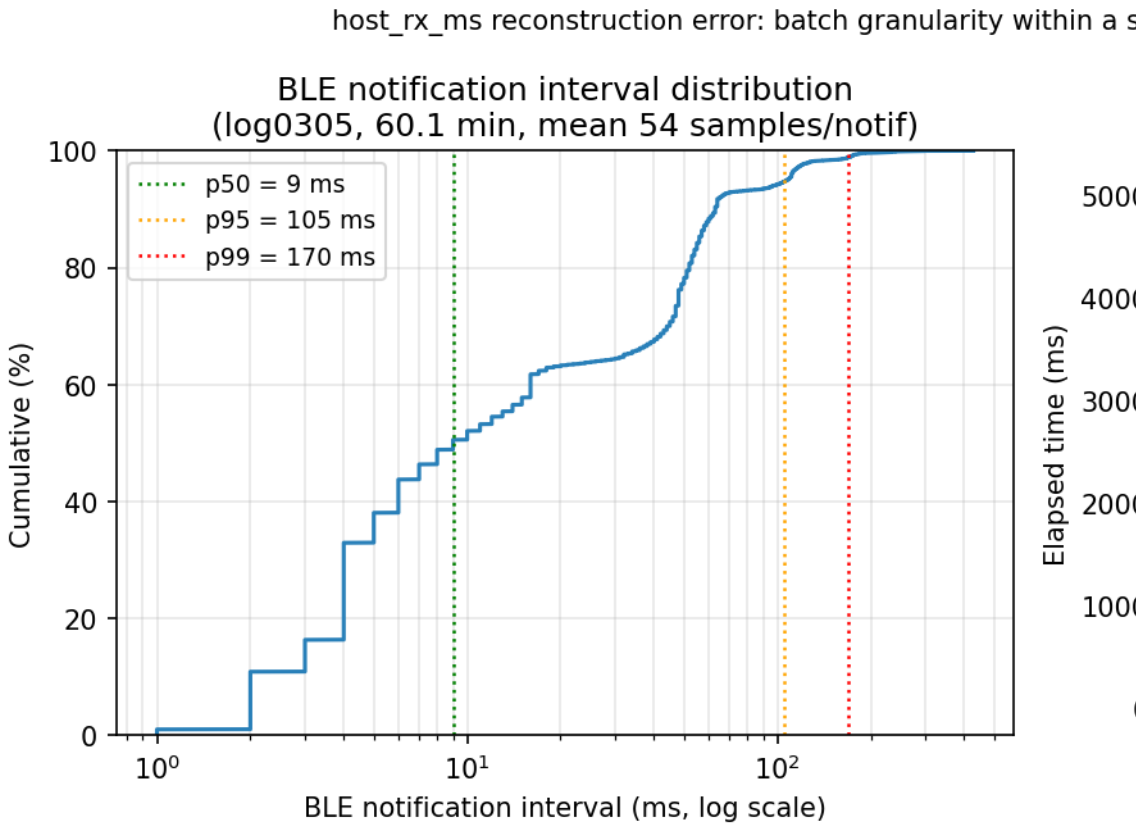


Figure 4. Fig. 4. abs_us reconstruction error characterization. Left: CDF of BLE notification intervals (log0305, 60.1 min). Right: 5-second window showing host_rx_ms batch granularity vs. dev_us per-sample precision.

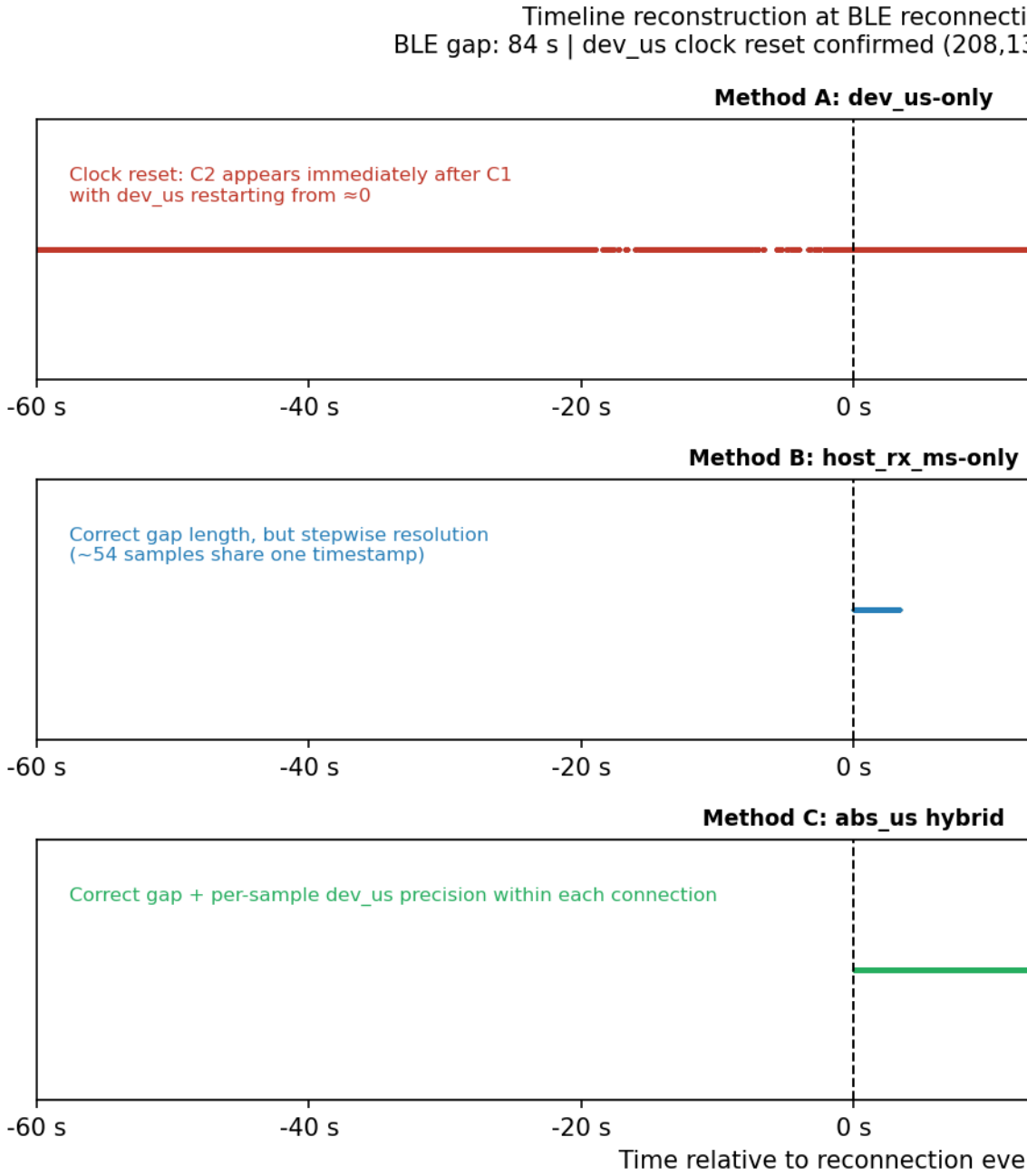


Figure 5. Fig. 5. Three-method timeline comparison at BLE reconnection (log0303). Top: Method A (dev_us) shows clock reset as zero-gap. Middle: Method B (host_rx_ms) shows correct gap with batch granularity. Bottom: Method C (abs_us hybrid) shows correct gap with per-sample precision.

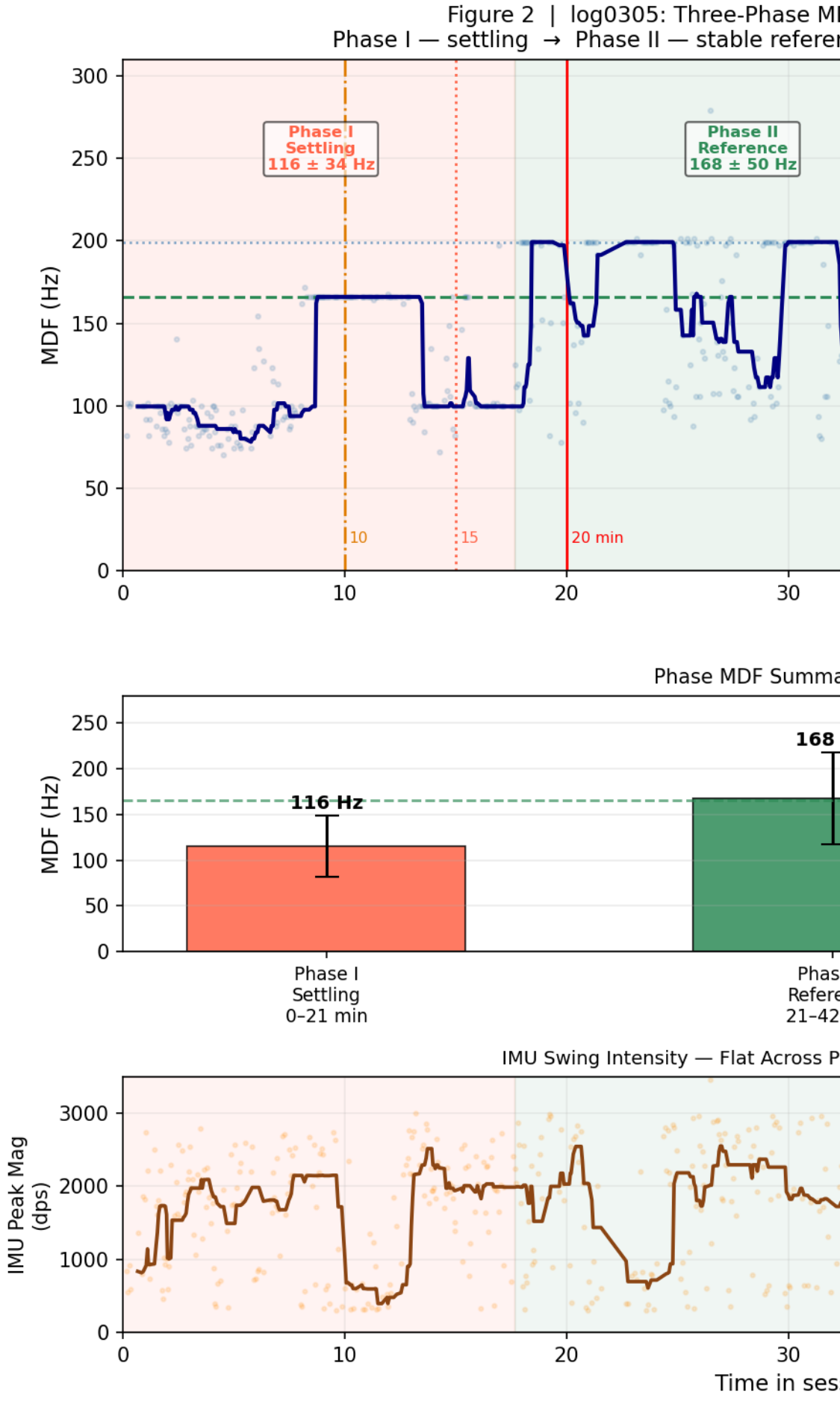


Figure 6. Fig. 6. Three-phase MDF trajectory in log0305 (63.1 min, 626 swings). Phase I: settling (0–21 min, MDF 115.7 ± 33.5 Hz). Phase II: stable reference (21–42 min, MDF 167.6 ± 50.1 Hz). Phase III: fatigue-consistent decline (42–63 min, MDF 149.5 ± 45.1 Hz). IMU remains flat across all phases (r =

https://doi.org/10.3390/s1010000

Figure 4 | Warm
Standard 10-min skip comple

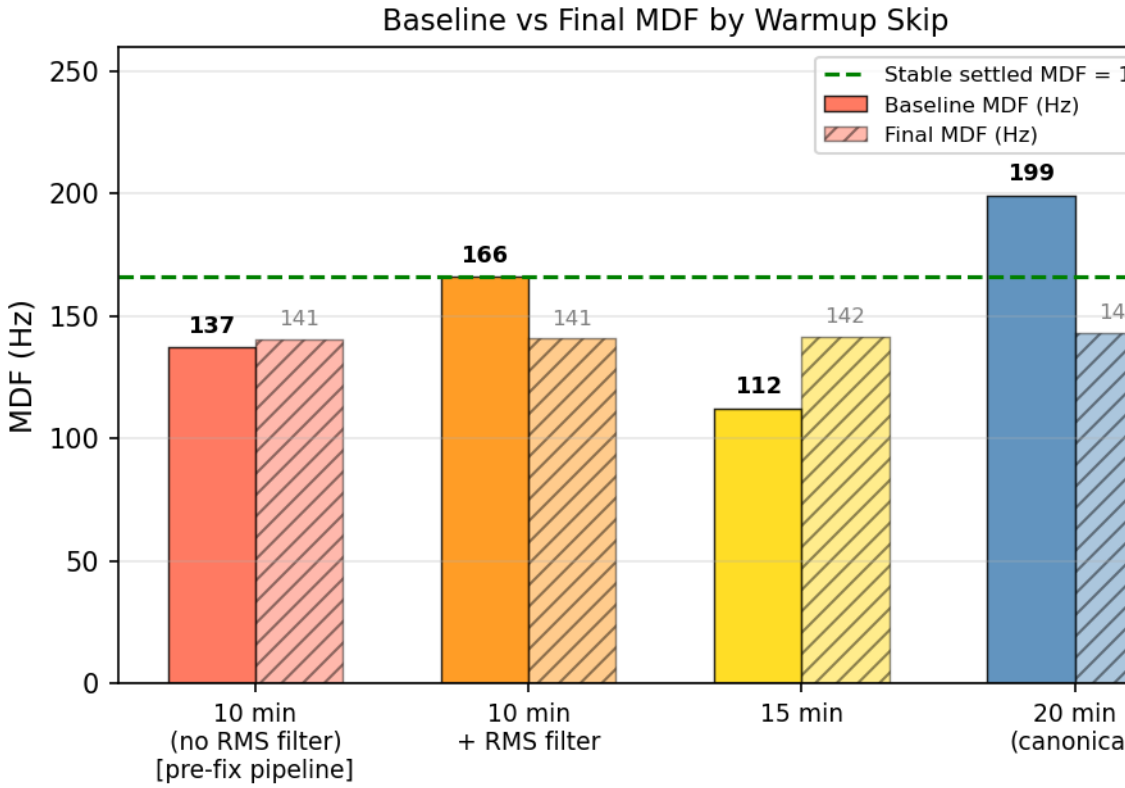


Figure 7. Fig. 7. Warmup-skip sensitivity. Left: baseline MDF per skip condition vs. stable settled reference. Right: MDF drop % per condition.

Phase I (settling): MDF 115.7 ± 33.5 Hz — substantially below the settled reference. Phase II (reference): MDF 167.6 ± 50.1 Hz — post-settling stable baseline. Phase III (fatigue-consistent): MDF 149.5 ± 45.1 Hz — decline after the settled reference.

The critical observation: early and late low-MDF regions have different causes. Early suppression reflects contact settling — MDF subsequently recovered while IMU swing intensity remained flat, consistent with electrode contact improvement rather than physiological fatigue. Late decline occurs after the settled reference and is fatigue-consistent.

6.2. 6.2 Warmup-Skip Sensitivity

Figure 7 shows that the same session gives completely different fatigue conclusions depending on warmup exclusion.

- 10-min skip: −2.6% MDF change (fatigue missed)
- 15-min skip: −26.2% (inverted — baseline still in settling phase)
- 20-min skip: +28.1% MDF decline, +17.3% MNF decline (fatigue-consistent result)

The non-monotone behavior (15-min worse than 10-min) reflects the RMS-based baseline selector catching swings from the steepest settling transition. These results demonstrate that warmup-skip sensitivity analysis is a necessary robustness audit for no-gel field sEMG fatigue studies.

6.3. 6.3 IMU-Conditioned Validity (log0305)

Figure 8 shows the IMU–EMG diagnostic matrix.

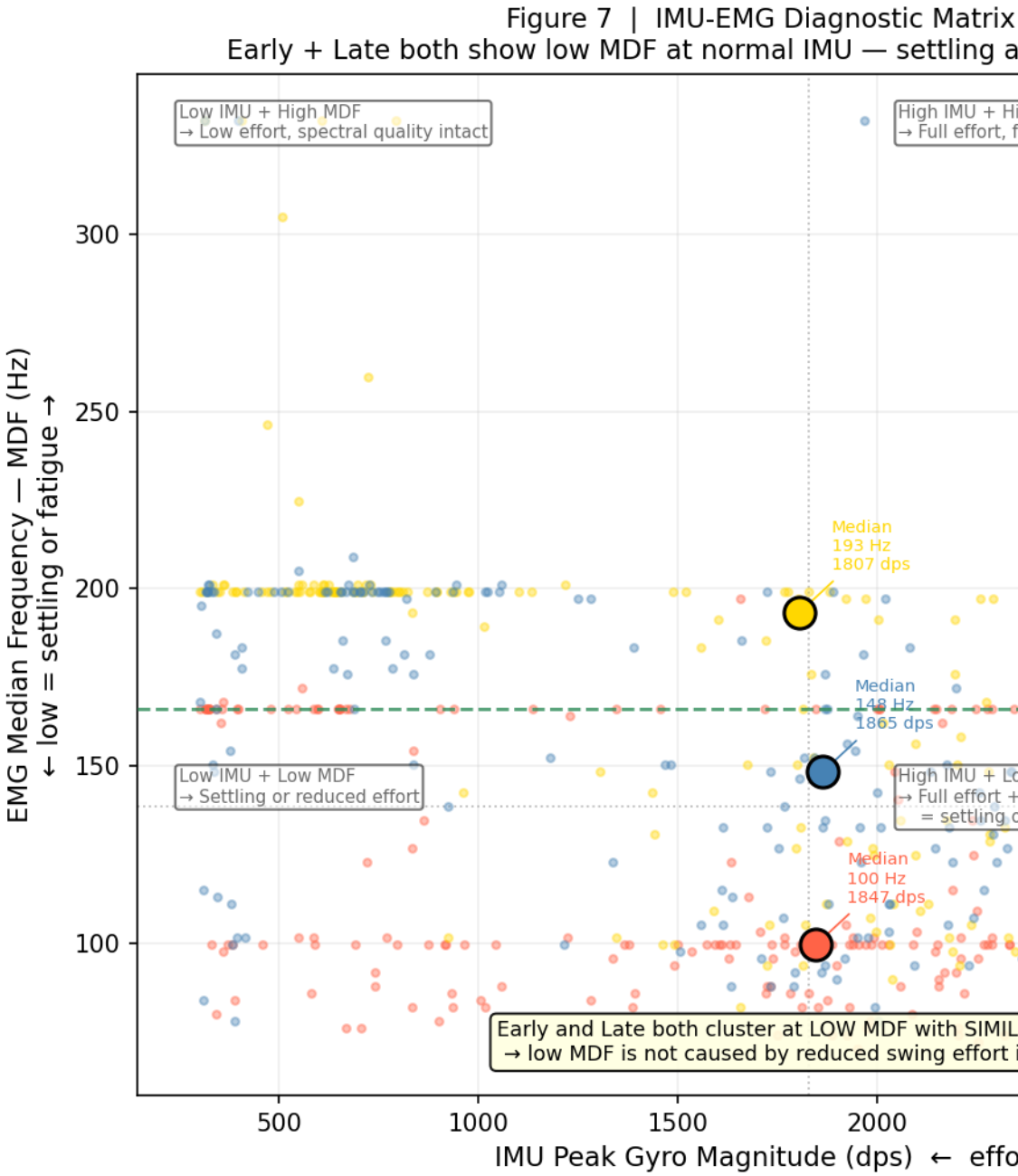


Figure 8. Fig. 8. IMU–EMG diagnostic matrix (log0305, 626 swings). IMU peak gyro magnitude vs. MDF, colored by session third. Early and Late swings cluster at lower MDF despite similar IMU intensity. r = +0.029, p = 0.468.

All three session thirds show similar IMU intensity (r = +0.029, p = 0.468), confirming that reduced gross swing kinematics is an unlikely explanation for either Phase I suppression or Phase III decline. Changes are attributed to electrode contact evolution (Phase I) and fatigue-consistent spectral shift (Phase III).

6.4. 6.4 Contact-Quality Evolution

In log0305, contact quality increased from early to mid-session (0.36 ± 0.28 to 0.49 ± 0.47) and declined late (0.25 ± 0.32), consistent with electrode settling followed by later activation-amplitude changes.

In log0303, contact quality differed substantially between segments (Segment 1: 0.26 vs. Segment 2: 0.59), supporting that the full-session MDF drop reflected a contact state change across the gap rather than a continuous physiological trend.

6.5. 6.5 Session-Gap Artifact (log0303)

Figure 10 shows the log0303 continuity failure mode.

The full-session 28.7% MDF drop compares Segment 1 against Segment 2 across a 66-min gap during which electrode contact state and player condition may have changed. Segment 1 alone shows 4.9% — below any meaningful fatigue threshold for 13.5 min of play. Treating disconnected BLE recordings as continuous physiological sessions is a real wearable failure mode requiring continuity auditing.

6.6. 6.6 NEW: Ball-Impact Masking Impact on MDF Estimates (log0305)

Figure 11 shows the per-swing MDF comparison between full-window and pre-impact window (Approach B).

Of 633 total swings, 512 (81%) showed ball impact (acc_dyn $\geq$ 3.0g, mean peak acc_dyn = 11.1g). Full-window MDF for impact swings: 133.7 ± 43.4 Hz (median 125.0 Hz). Pre-impact MDF (Approach B, [t − 600 ms, t − 20 ms]): 160.4 ± 52.0 Hz (median 168.0 Hz). Delta (pre-impact − full): mean = +26.7 Hz, std = 44.5 Hz, median = +3.9 Hz. Distribution: p5 = −8.7 Hz, p75 = +46.9 Hz, p95 = +101.6 Hz — right-skewed, with 25% of swings showing no shift and 50% showing positive shift.

Interpretation. The pre-impact window captures the active muscle contraction phase before mechanical disturbance, resulting in systematically higher MDF (less apparent fatigue effect) vs. the full window. The magnitude of the shift (+26.7 Hz mean) confirms that window selection is physiologically meaningful for impact-bearing swings and that the IMU enables principled temporal partitioning of EMG analysis windows.

6.7. 6.7 NEW: Multi-Subject Results (N3 — TODO)

[PLACEHOLDER — awaiting multi-subject data collection]

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7. 7. Discussion

7.1. 7.1 Platform–Methodology Coupling

The platform and methodology are tightly coupled — neither is useful without the other. The spectral fatigue analysis (§6.1–6.6) requires: - **RAW EMG access** — not available through the pogo-pin path; required the solder bypass (§5.5) - **abs_us timeline** — required for correct session-gap artifact detection (§6.5); Method A (dev_us-only) would have silently merged the two log0303 segments - **IMU-as-context** — required for swing detection, validity conditioning (§6.3), and pre-impact window masking (§6.6) - **2000 Hz EMG** — required for 50–450 Hz bandpass and Welch PSD; 200 Hz (Athos) is insufficient

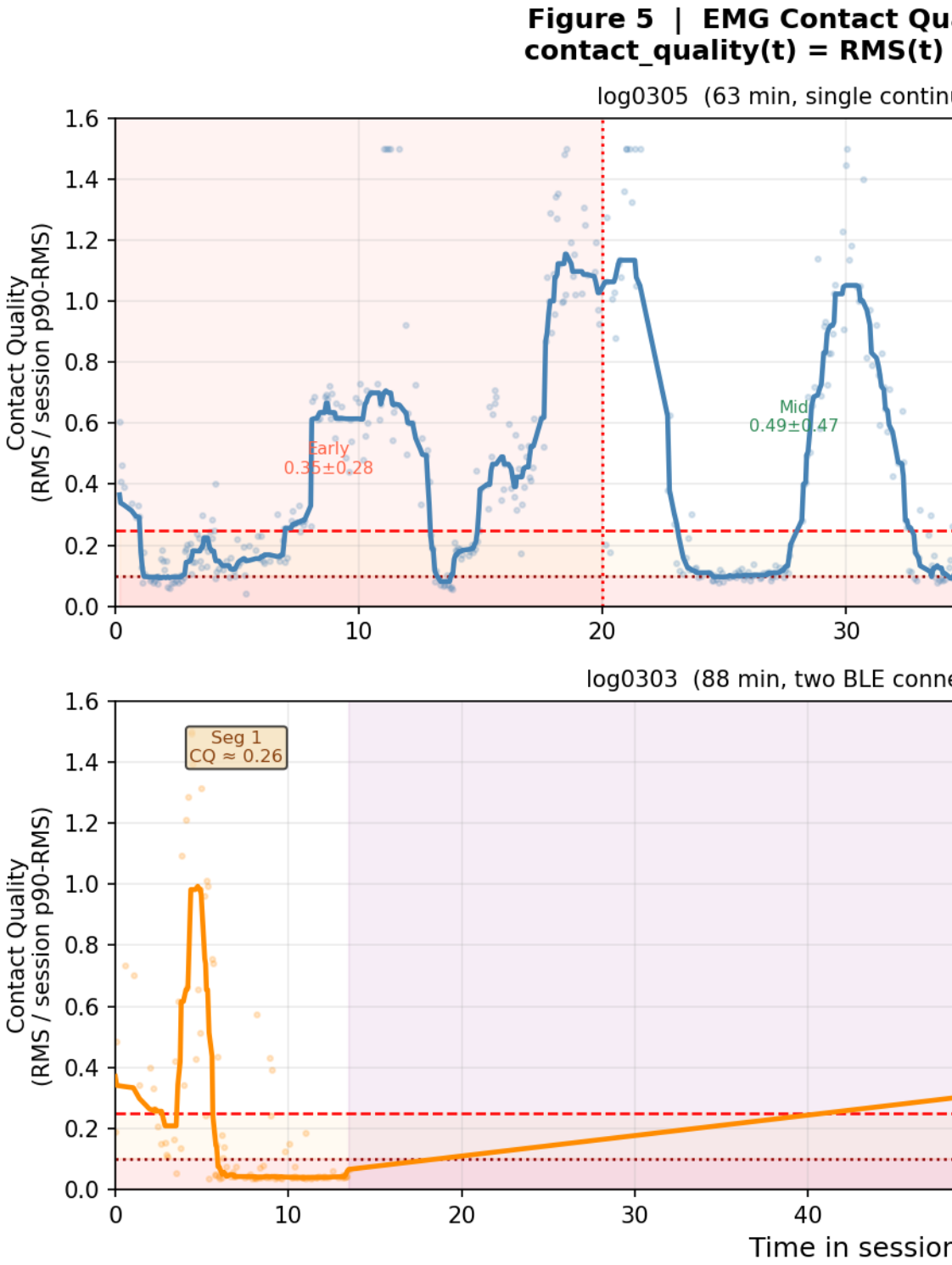


Figure 9. Fig. 9. Contact quality evolution (log0305).

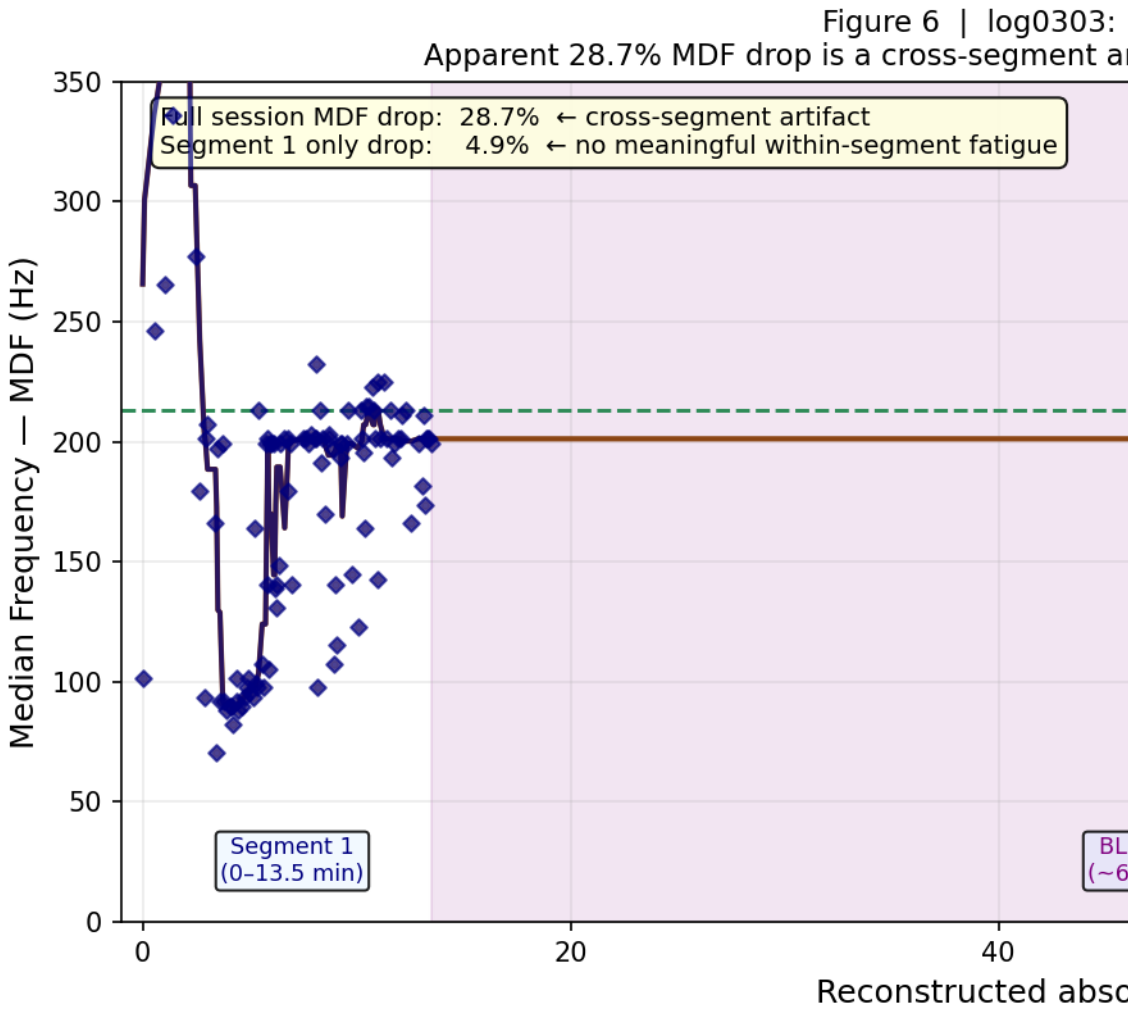


Figure 10. Fig. 10. log0303 gap artifact. Full session: 28.7% MDF drop. First continuous segment alone: 4.9% change. A 84-second BLE reconnection gap separates the two segments (66-min apparent gap in practice time).

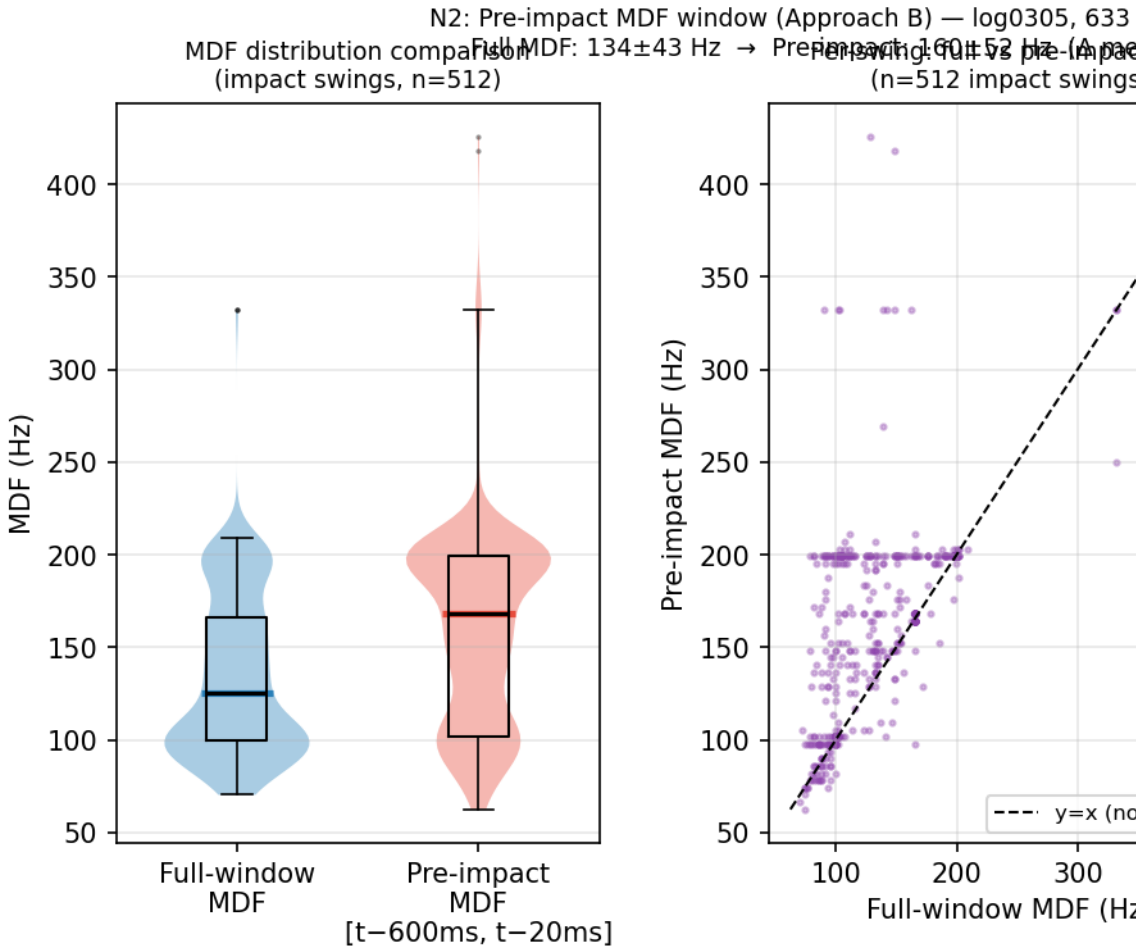


Figure 11. Fig. 11. Pre-impact MDF window analysis (log0305, 633 swings, 512 with impact $\geq$ 3.0g). Left: violin + box comparison. Center: per-swing scatter. Right: CDF of Δ MDF.

These dependencies motivate the combined paper: a platform paper without the methodology would not demonstrate why ±25 μs timestamps and RAW EMG access matter; a methodology paper without the platform would be unable to justify the novel contributions as field-deployable.

7.2. 7.2 Three-Phase Trajectory as a General Framework

The three-phase settling–reference–fatigue trajectory is predicted by dry-electrode physics (§4.2) and should appear in any no-gel field sEMG deployment with sufficient session length. Phase I duration depends on sweat rate and ambient conditions; in hot weather or high-effort sport, settling may be shorter. In air-conditioned environments or low-activity segments, settling may persist beyond 20 min. The warmup-skip sensitivity protocol (§4.6, §6.2) provides a session-specific robustness check regardless of phase boundaries.

7.3. 7.3 Pogo-Pin Failure Mode Generalization

ADC multiplexer charge sharing is not specific to MyoWare or pogo pins. Any wearable routing a high-impedance analog signal through a non-soldered contact and sampling on a multiplexed ADC is vulnerable. The diagnostic heuristic ($r > 0.95$ between independent channels, near-zero variance on one, post-filter flatline) generalizes to capacitor microphones, piezo sensors, and other high-impedance biosignal sources routed through multiplexed ADCs.

7.4. 7.4 Practical Recommendations for Field sEMG Deployment

1. **Use ≥20 min warmup exclusion** for no-gel forearm snap-electrode deployments
2. **Run warmup-skip sensitivity** as a standard robustness audit (report effect of 10/15/20 min)
3. **Use IMU swing intensity** as an independent validity channel, not just a motion feature
4. **Audit BLE session continuity** before computing cross-session baselines
5. **Verify RAW EMG access** via pogo-pin / connector inspection before field deployment
6. **Use pre-impact MDF windows** when IMU-detected ball contact events contaminate windows

7.5. 7.5 Limitations

- Single subject for currently reported results (n=1 pilot); multi-subject generalization is N3 (**TODO**)
- No MVC, RPE, grip-force, or ball-speed ground truth — findings are fatigue-consistent rather than physiologically confirmed
- Settling duration varies with sweat rate, ambient conditions, and electrode state; 20-min skip is empirically selected for this session, not a universal threshold
- No direct in-session impedance measurement — settling is inferred indirectly from MDF/RMS
- BLE 4.2 throughput ceiling: combined data rate (~12.5 KB/s) is near MTU 244 capacity; adding more sensors requires rate reduction or BLE 5.0 migration
- Contact quality metric not validated against direct impedance

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8. 8. Conclusion

The platform–methodology coupling demonstrated here enables field sEMG fatigue monitoring that would not be possible with either component alone. The $99 ESP32

wearable provides: ±25 μs timestamp accuracy (99.03% over 2.3M samples), self-describing TLV protocol surviving three FSR revisions, zero detected sample drops during 150+ min of field sessions, and documented pogo-pin failure diagnosis and repair.

The methodology reveals that no-gel field tennis sEMG fatigue monitoring is fundamentally a trustworthiness problem. The same session changed from −2.6% to +28.1% MDF drop depending solely on the warmup exclusion window, demonstrating that baseline selection is a first-order determinant of fatigue interpretation. The IMU, as an independent kinematic context channel, confirmed stable swing intensity (r = +0.029, p = 0.468) across all phases, attributing Phase I suppression to electrode settling and Phase III decline to fatigue-consistent spectral shift. A 84-second BLE reconnection gap created an apparent 28.7% fatigue signal that disappeared to 4.9% for the first continuous segment — demonstrating the necessity of continuity auditing.

Pre-impact MDF windowing (Approach B) using IMU-detected ball contact shows a mean shift of +26.7 Hz vs. full-window MDF, confirming that temporal partitioning guided by IMU events is physiologically meaningful for impact-bearing swings.

These findings establish a trustworthiness-first framework for wearable sEMG fatigue monitoring in field sports deployment, applicable beyond tennis to any dynamic sport requiring rapid electrode preparation and wireless continuous monitoring.

9. 9. References

10. Appendix A: Figure Inventory

Journal Fig	Source file	Caption source
1	icpics_figures/fig_system_architecture.png	§3.2
2	icpics_figures/fig_timing_accuracy.png	§3.3
3	icpics_figures/fig_protocol.png	§3.4
4	paper_figures/fig_absus_error_cdf.png	§5.4 (N1)
5	paper_figures/fig_absus_timeline_methods.png	§5.4 (N1)
6	paper_figures/fig2_electrode_settling.png	§6.1
7	paper_figures/fig4_warmup_skip_sensitivity.png	§6.2
8	paper_figures/fig7_imu_emg_diagnostic_matrix.png	§6.3
9	paper_figures/fig5_contact_quality.png	§6.4
10	paper_figures/fig6_log0303_gap_artifact.png	§6.5
11	paper_figures/fig_n2_preimpact_mdf.png	§6.6 (N2)
12	[TBD — N3 cross-subject figure]	§6.7 (N3 TODO)

11. Appendix B: Reproducibility Notes

All figures in this paper can be regenerated from recorded analysis-output directories: - N1 (abs_us validation): `python abs_us_validate.py` → `paper_figures/fig_absus_*.png` - N2 (pre-impact MDF): `python n2_preimpact_mdf.py` → `paper_figures/fig_n2_preimpact_mdf` - §6.1–6.5 figures: `python paper_figures_runner.py` (TODO: create this script) - N3 (multi-subject): awaiting data collection

LaTeX build (MDPI Sensors format):

```
cd mdpi_template && pdflatex journal_sensors.tex && pdflatex journal_sensors.tex
```

Two passes required: first pass compiles content; second pass resolves cross-references and page numbers. Output: `mdpi_template/journal_sensors.pdf`.

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